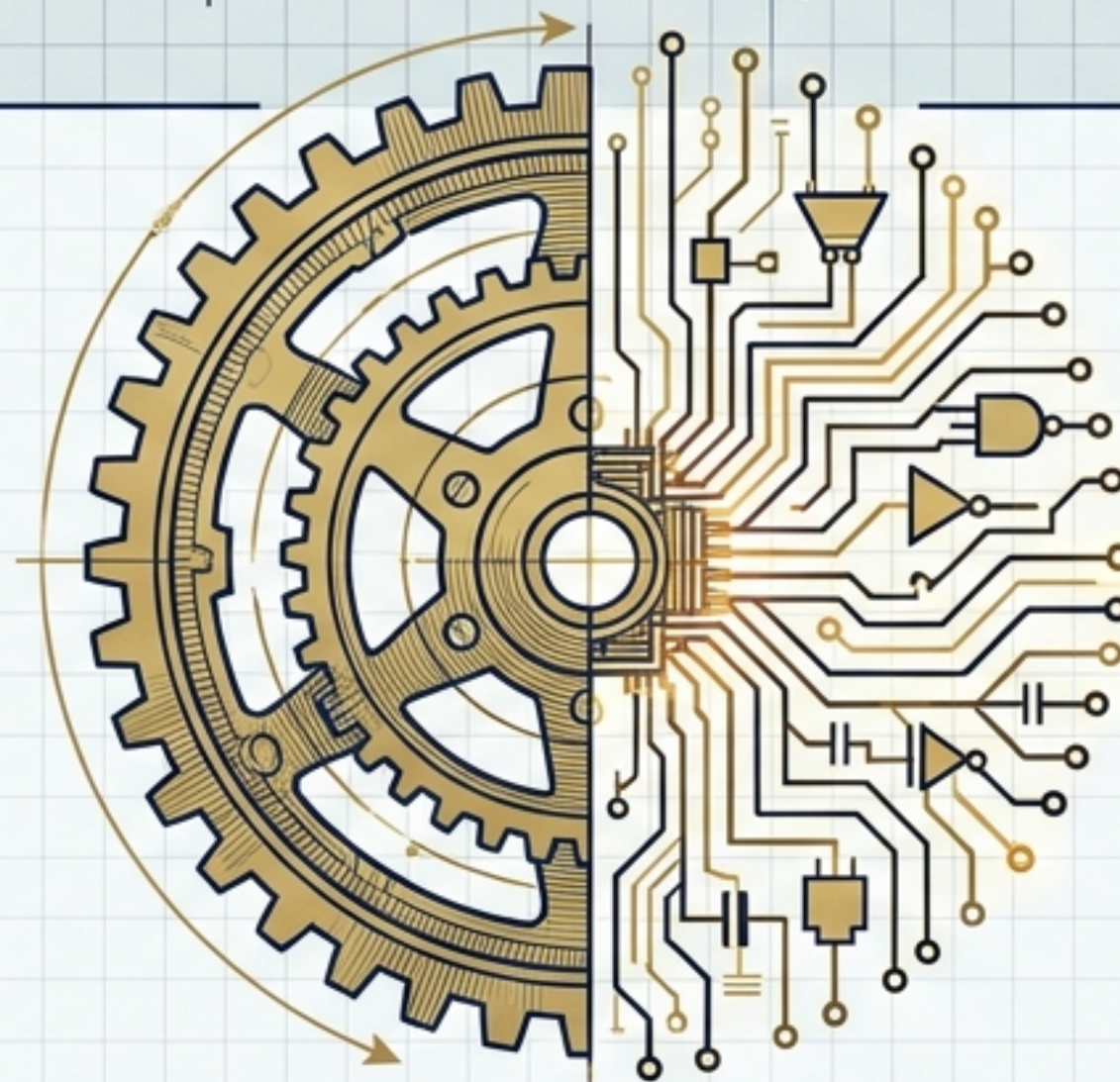


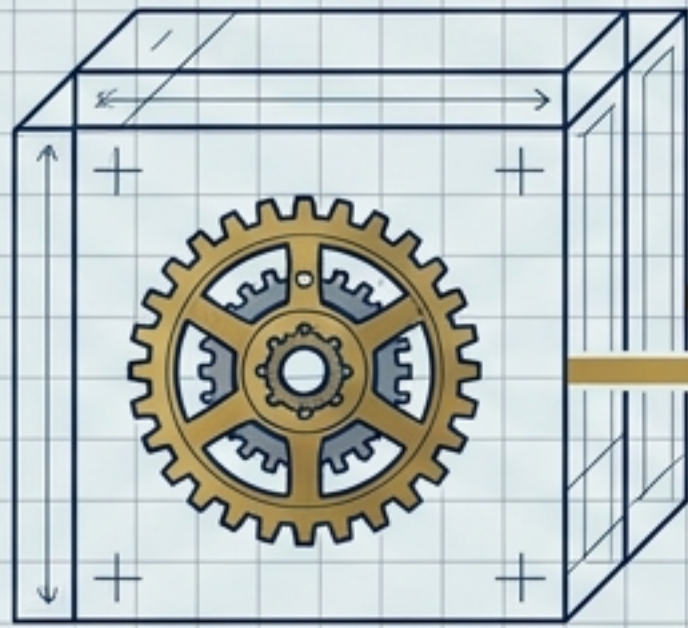
[FILE NO. 1843-1957]
[STATUS: DECLASSIFIED]
[SUBJECT: THE FOUR PILLARS OF MODERN COMPUTING]



The Architecture of the Information Age

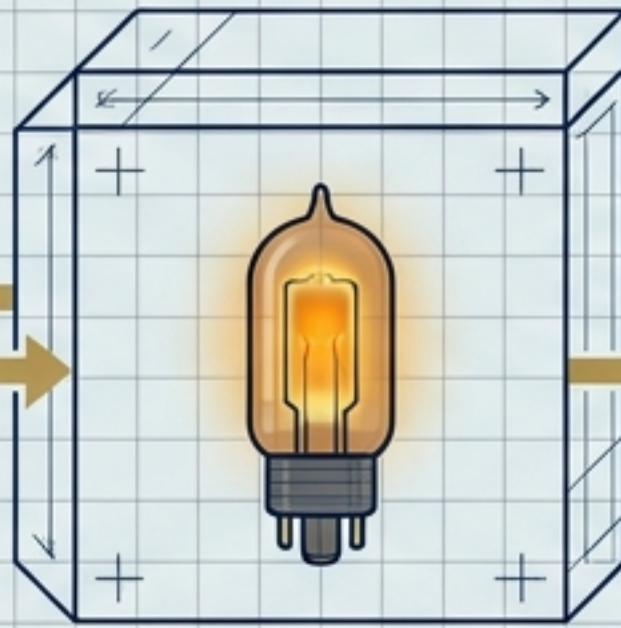
From Brass Logic to Silicon Scale

THE FOUR PILLARS OF THE INFORMATION AGE



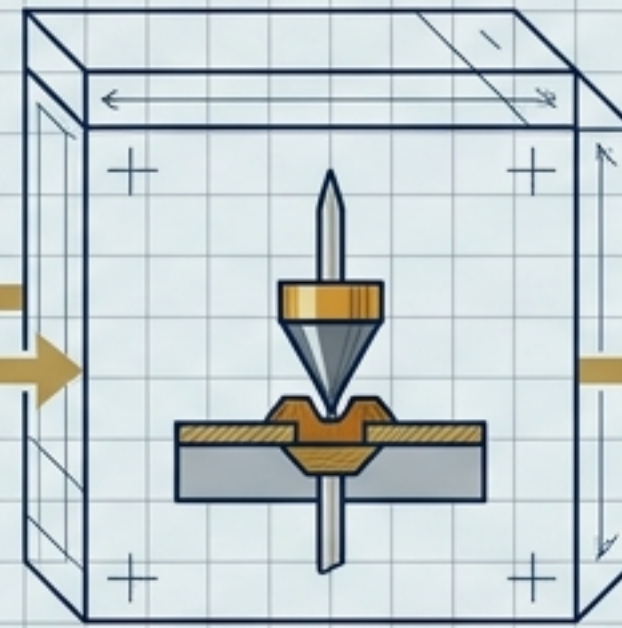
THE LOGIC (1843)

Separating logic from arithmetic.
Key figures: Babbage & Lovelace.



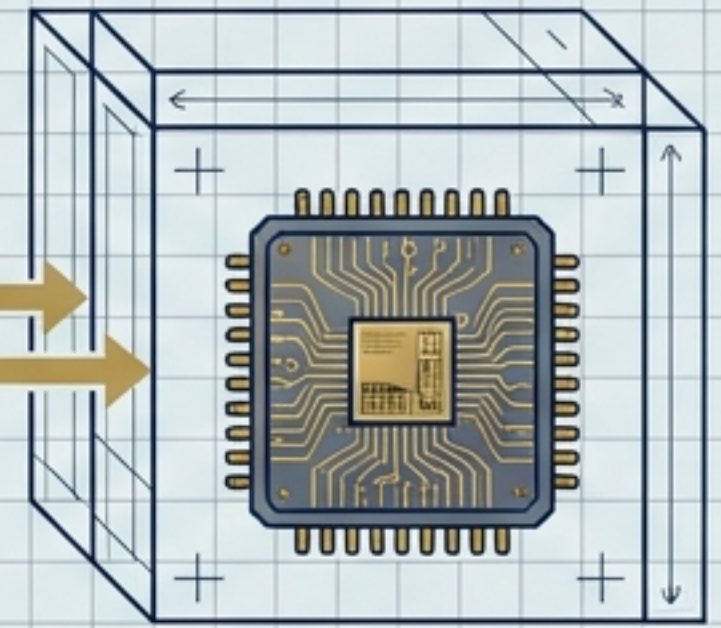
THE MACHINE (1946)

The operational dawn of electronic computing.
Key artifact: ENIAC.



THE SWITCH (1947)

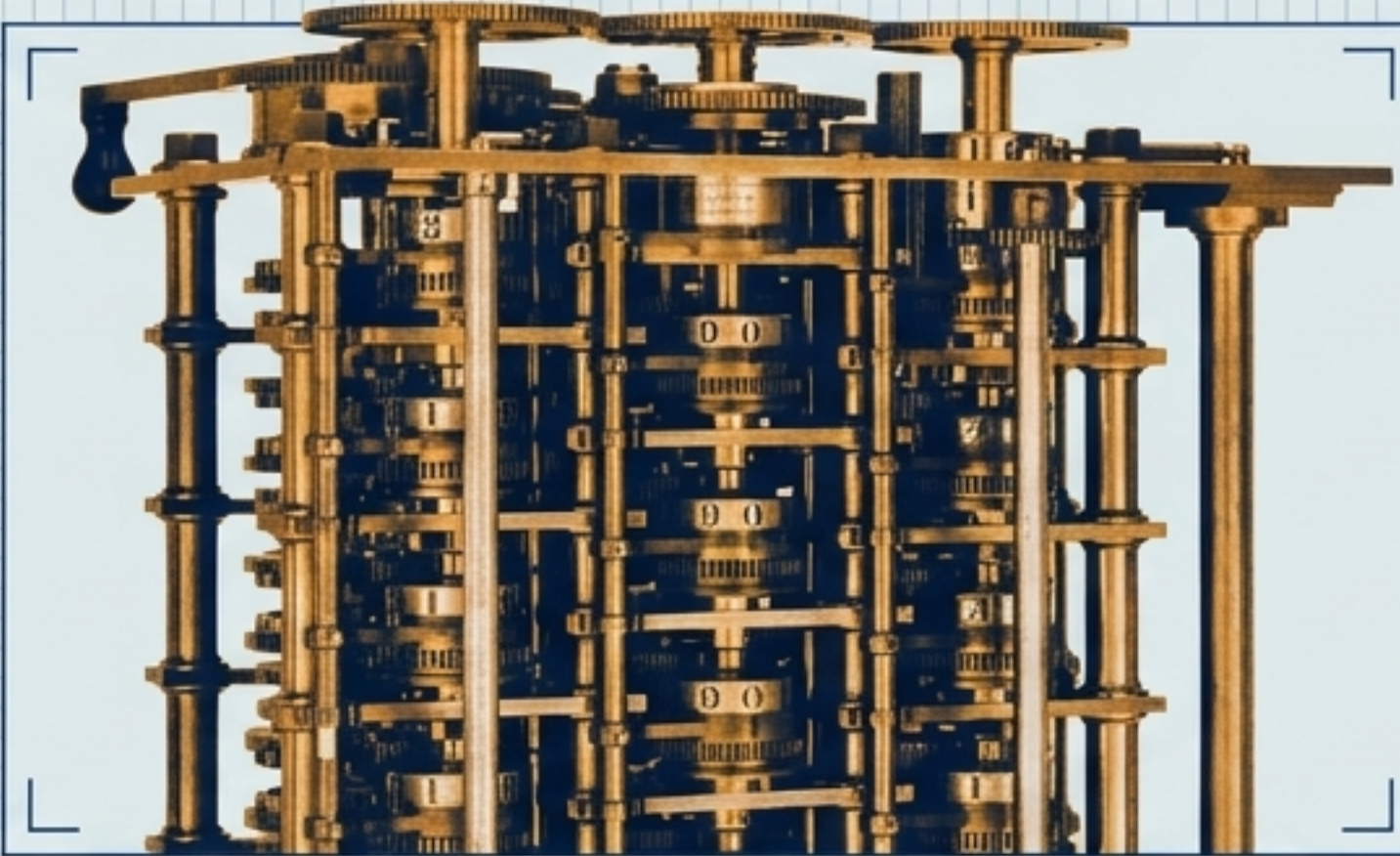
Overcoming physical limits with solid-state physics.
Key artifact: Point-Contact Transistor.



THE SCALE (1957)

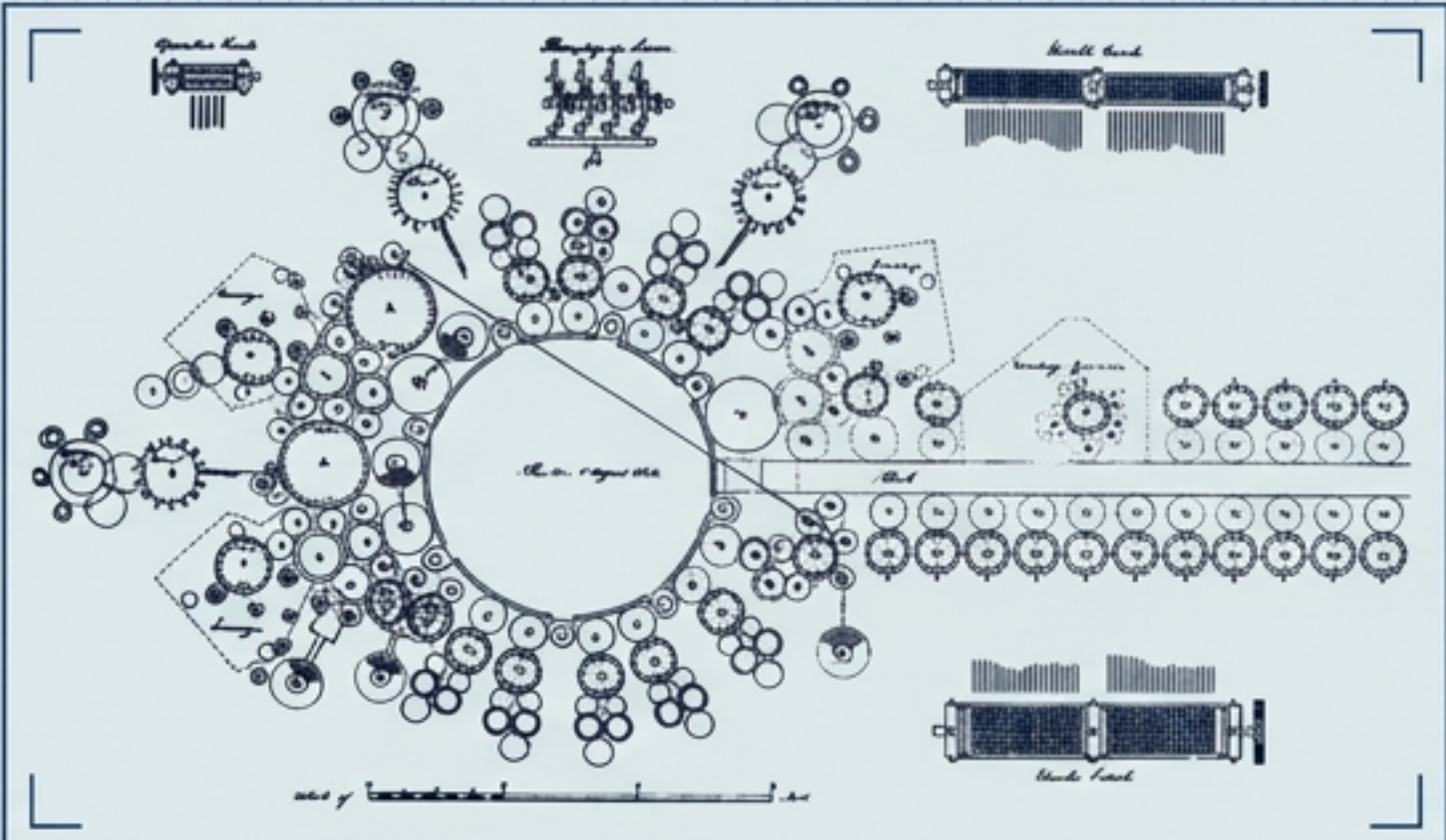
Shrinking and integrating for mass manufacturing.
Key artifact: Integrated Circuit.

Each era changed the medium, but merely amplified the scale of the original logic.



Difference Engine:

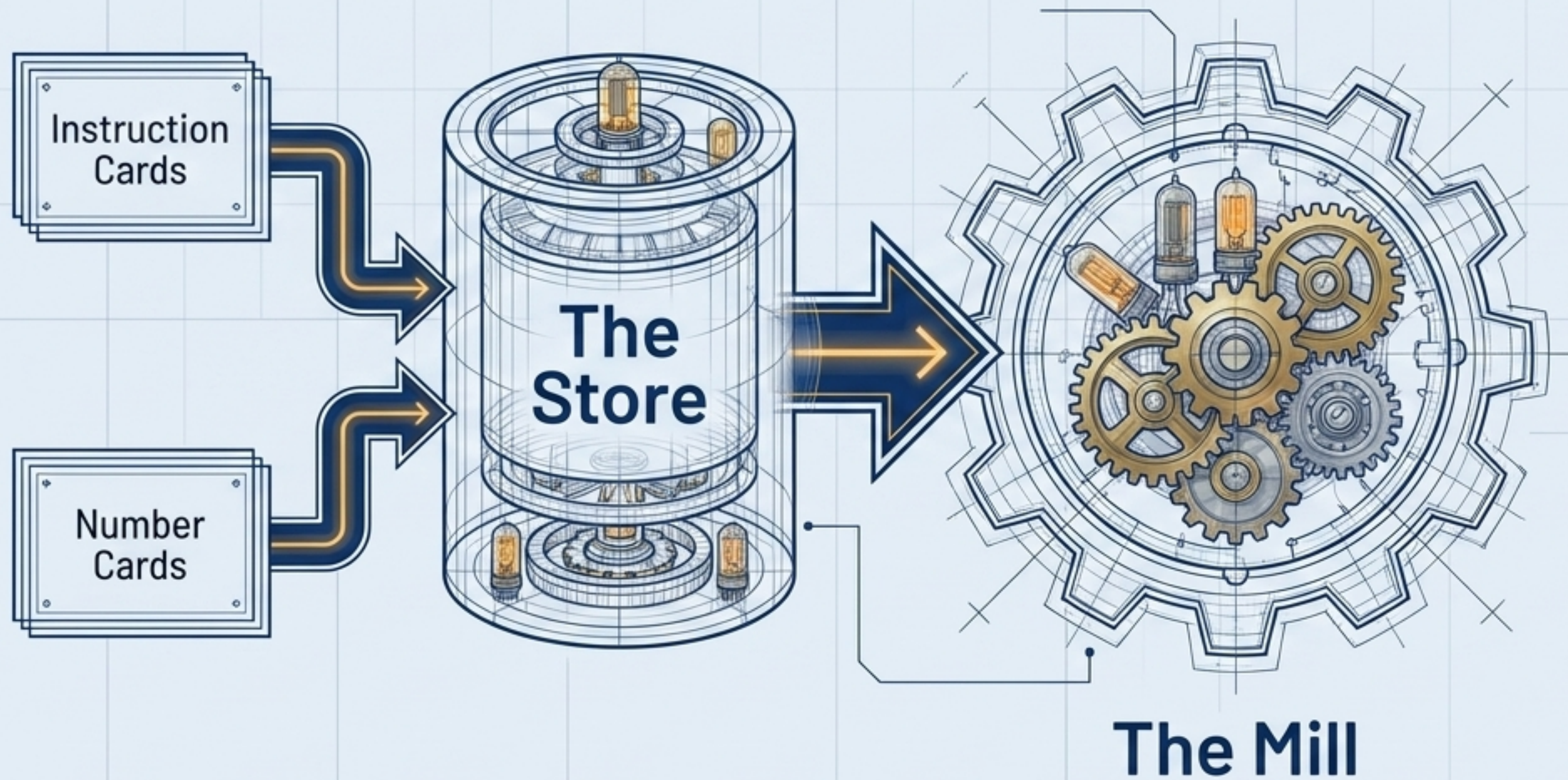
- Plane: Logically lower.
- Operation: Arithmetical (+, -, ×, ÷).
- Iteration: Finite / Primitively Recursive.



Analytical Engine:

- Plane: Logically universal.
- Operation: Programs directed via punched cards.
- Iteration: Conditional Iteration (looping until a test succeeds).

The bounds of arithmetic were outstepped the moment of applying the cards had occurred. — Ada Lovelace



The Breakthrough

Programs instructions are coded as numbers. A universal machine can decode and simulate them.

The Missing Link

Babbage and Lovelace lacked the formal mathematical coding methods (later developed by Gödel and Turing in the 1930s) to fully map this, but they understood the vast potential.

Lovelace's Vision

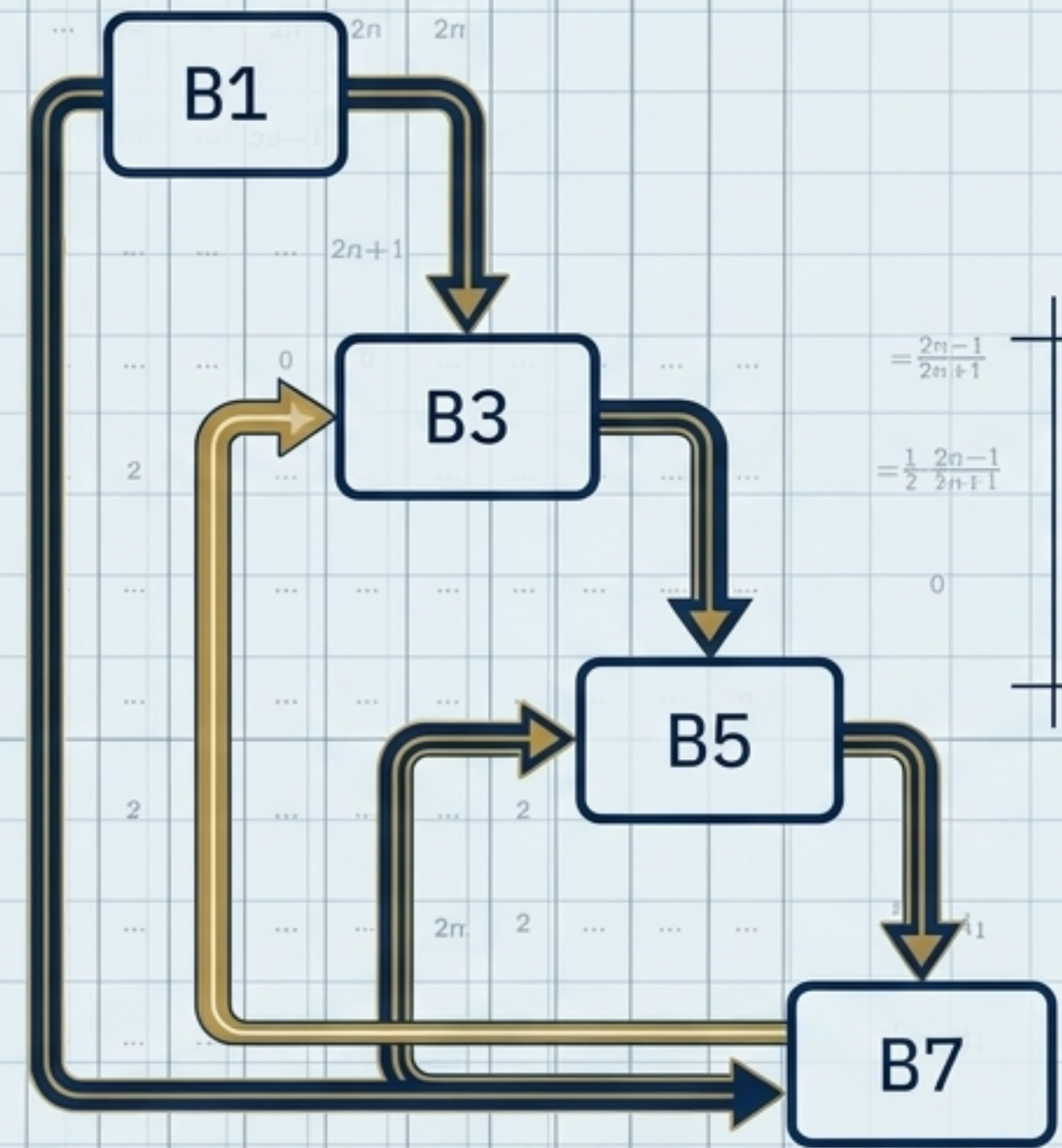
The engine might compose elaborate and scientific pieces of music of any degree of complexity or extent.

Programming Memory: The Bernoulli Sequence

Data.										Working Variables.			Result Variables.			
1V_1	1V_2	1V_3	0V_4	0V_5	0V_6	0V_7	0V_8	0V_9	${}^0V_{10}$	${}^9V_{11}$	${}^9V_{12}$	${}^9V_{13}$	${}^1V_{21}$	${}^1V_{22}$	${}^1V_{23}$	${}^1V_{24}$
○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○
0	0	0	0	0	0	0	0	0	0	0	0	0	B_1 in a decimal fraction.	B_3 in a decimal fraction.	B_5 in a decimal fraction.	0
1	2	4	0	0	0	0	0	0	0	0	0	0	B_1	B_3	B_5	0
1	2	n														

1	$\times$	${}^1V_2 \times {}^1V_3$	${}^1V_4, {}^1V_5, {}^1V_0$	$\begin{cases} {}^1V_2 = {}^1V_2 \\ {}^1V_3 = {}^1V_3 \end{cases}$	$= 2n$
2	$-$	${}^1V_4 - {}^1V_1$	1V_4	$\begin{cases} {}^1V_2 = {}^2V_3 \\ {}^1V_5 = {}^1V_4 \end{cases}$	$= 2n-1$
3	$+$	${}^1V_5 + {}^1V_7$	2V_5	$\begin{cases} {}^1V_5 = {}^2V_5 \\ {}^1V_1 = {}^1V_1 \end{cases}$	$= 2n+1$
4	$+$	${}^2V_5 +$			
5	$+$	${}^1V_{12} +$			
6	$-$	${}^0V_{13} +$			
7	$-$	${}^0V_3 - {}^1V_1$	${}^1V_{10}$	$\begin{cases} {}^1V_2 = {}^2V_0 \end{cases}$	$= n-1(-3)$
8	$+$	${}^1V_2 + {}^0V_7$	1V_7	$\begin{cases} {}^1V_2 = {}^1V_2 \\ {}^0V_7 = {}^1V_7 \end{cases}$	$= 2+0=2$
9	$+$	${}^1V_6 + {}^1V_7$	${}^9V_{11}$	$\begin{cases} {}^1V_2 = {}^1V_8 \\ {}^9V_{11} = {}^2V_{11} \end{cases}$	$= \frac{2n}{5} = A_1$
10	$\times$	${}^1V_{12} \times {}^3V_{11}$	${}^1V_{12}$	$\begin{cases} {}^1V_{21} = {}^1V_{21} \\ {}^2V_{11} = {}^2V_{11} \end{cases}$	$= B_1, \frac{2n}{2} = B_1 A_1$
11	$+$	${}^1V_{12} + {}^1V_{12}$	${}^2V_{12}$	$\begin{cases} {}^1V_{12} = {}^3V_{12} \\ {}^1V_{10} = {}^2V_{13} \end{cases}$	$= -\frac{1}{2} \cdot \frac{2n-1}{3n+1} + B_1 \cdot \frac{2n}{3}$
12	$-$	${}^1V_{10} - {}^1V_7$	${}^2V_{10}$	$\begin{cases} {}^1V_{10} = {}^2V_{10} \\ {}^1V_1 = {}^1V_1 \end{cases}$	$= n-2=2$

Definition: Course-of-Values Recursion requires the complete course of all previous values to get the next one, forcing the machine to store, retrieve, and loop data.



Significance: This is not just a calculation; it is an engineered set of operational instructions—the birth of software.

Project PX (ENIAC): Unveiled February 14, 1946 at Moore School of Engineering

Physical Footprint: 30 Tons
// 1,500 Square Feet

Components: 17,000 Vacuum Tubes //
70,000 Resistors // 6,000 Manual Switches

Power Consumption:
174,000 Watts



Added 5,000 numbers in 1 second. Calculated a 30-second artillery shell trajectory in just 20 seconds—faster than the shell itself could fly.

Originally built to calculate ballistic trajectories for WWII, its first actual test was evaluating feasibility designs for the H-bomb in 1945.

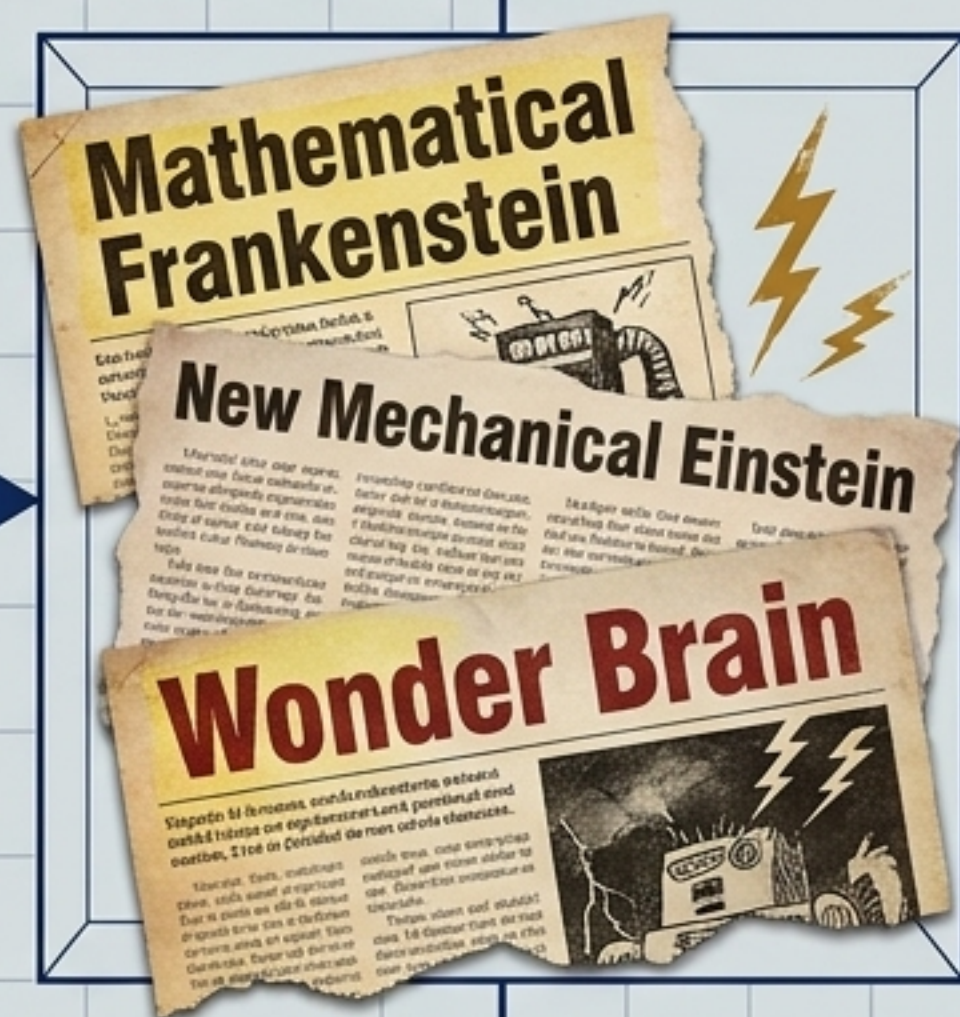
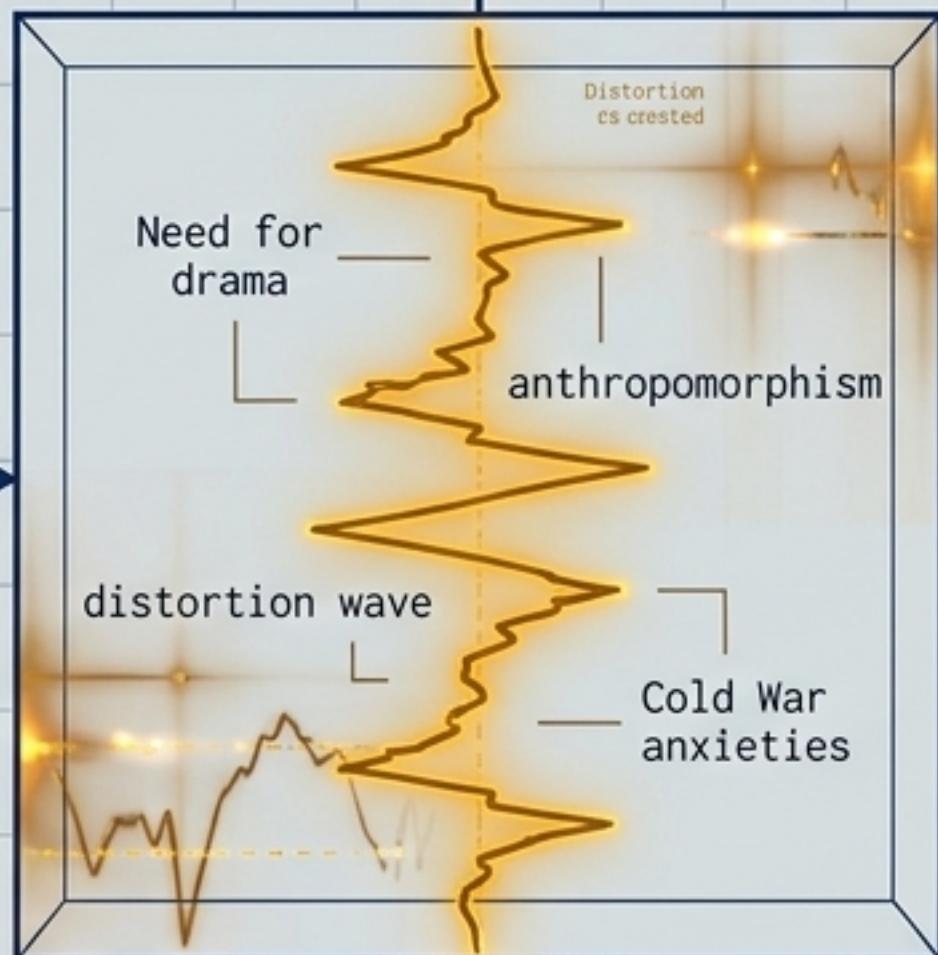
[INPUT]
Engineering Reality

[FILTER]
The Media Filter

[OUTPUT]
Public Perception

A calculating machine doesn't know how to do algebra, but only arithmetic. It does not replace original human thinking.

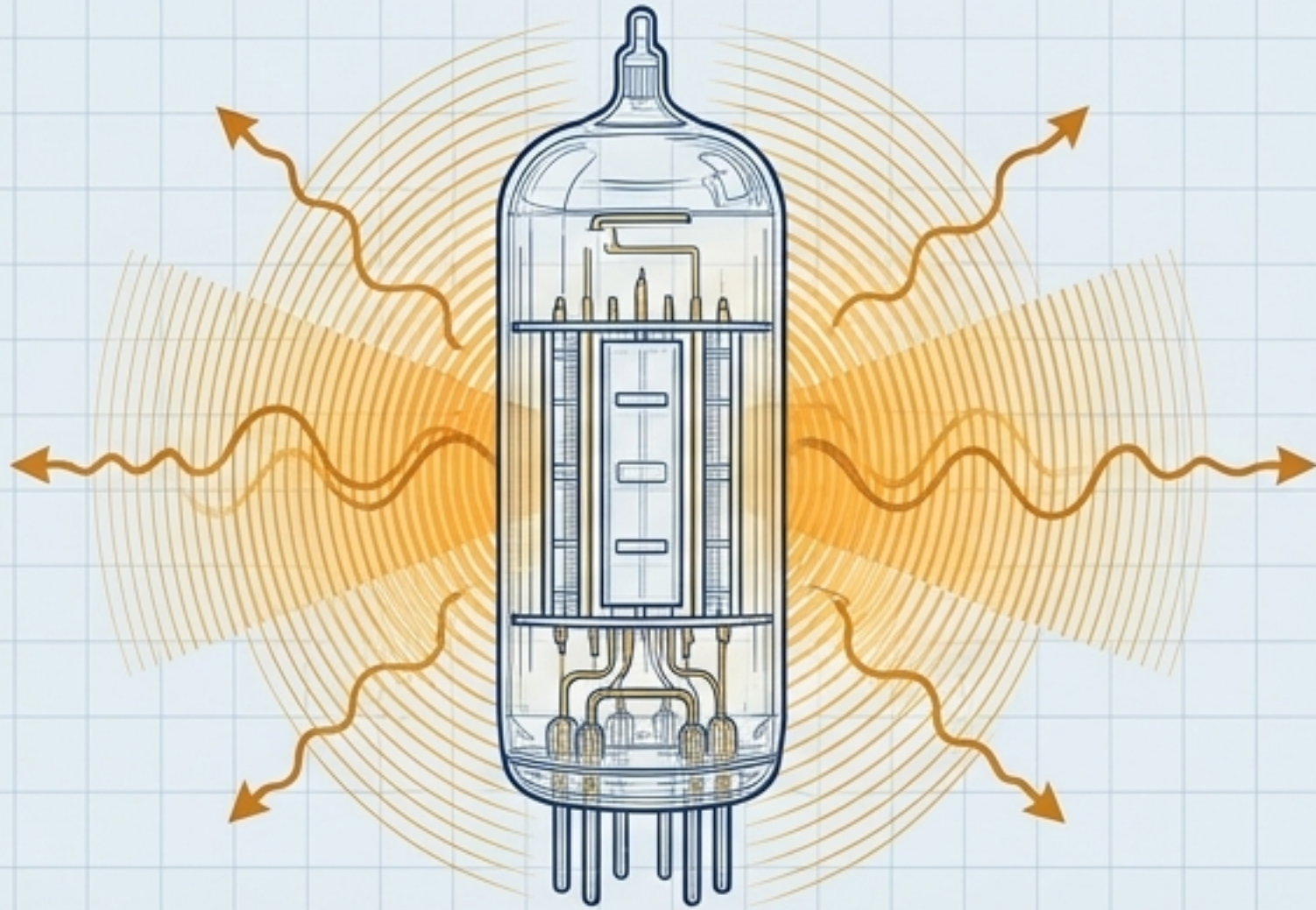
— John Mauchly



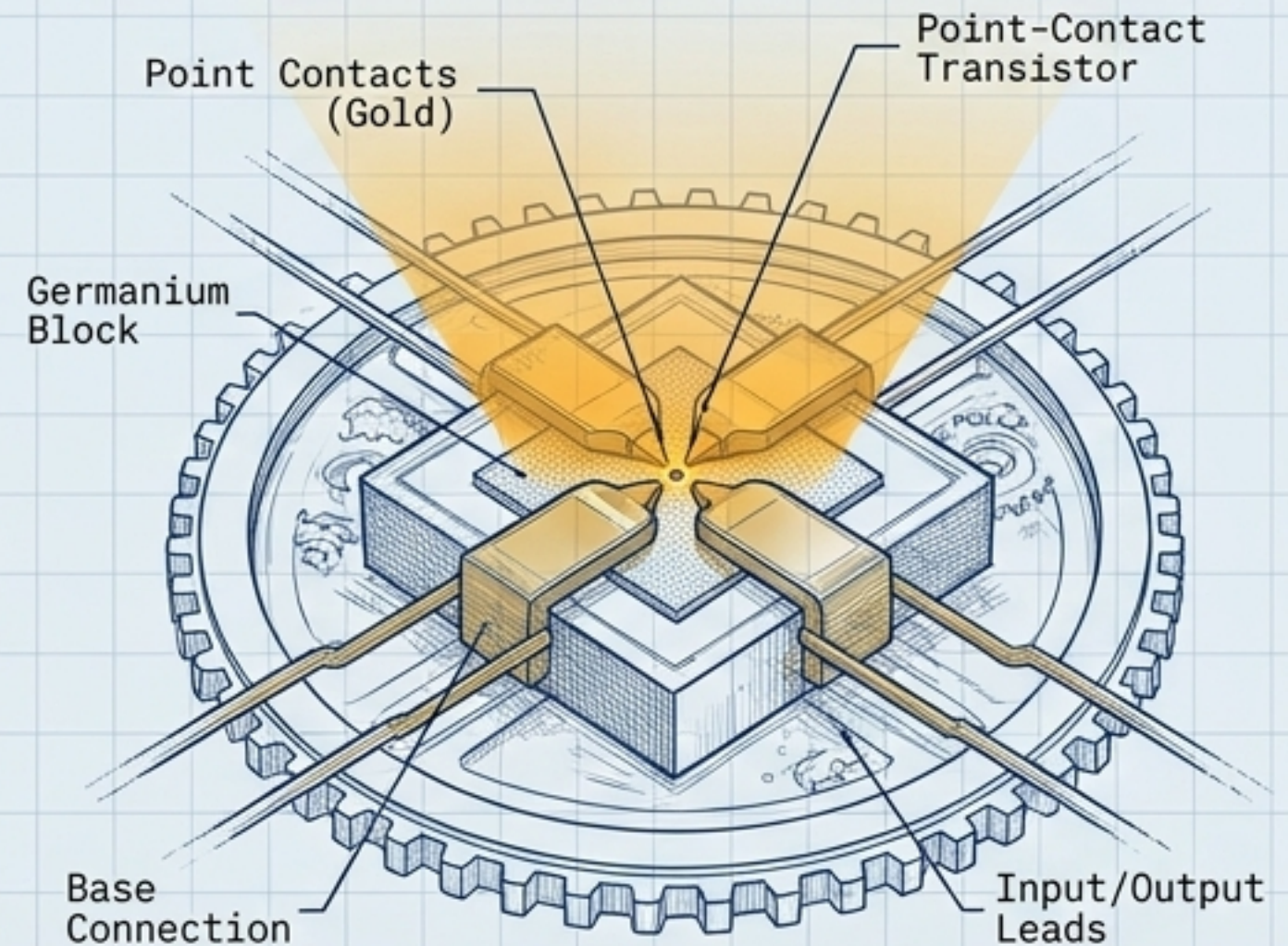
Result: The permanent cultural shift from computer as a human calculating to computer as an awesome, frightening electronic brain.



The Switch: The Miracle Month of 1947



The Problem: Vacuum tubes were massive, costly, and emitted crippling amounts of heat.



The Solution: December 16, 1947 (Bell Labs) — Bardeen, Shockley, and Brattain fabricate the first point-contact transistor using semiconducting germanium.

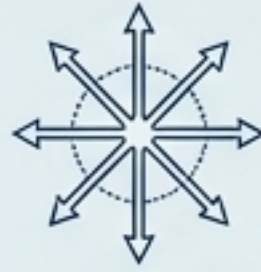
The Impact: Amplified input signals by 100x without the heat or fragility of glass tubes, ushering in the modern era of solid-state electronics.

The Scale: Fairchild and the Planar Process

[The Defection (Sept 1957)]

The Traitorous Eight leave Shockley to found Fairchild Semiconductor, planting the seeds of Silicon Valley.

DATA: MONON, Sept, 1957



Exodus to Fairchild

[Hoerni's Planar Process (1959)]

Covering the transistor with a flat insulating layer of silicon dioxide (glass), transforming semiconductor production from fragile, handcrafted 3D structures to high-volume automated printing.

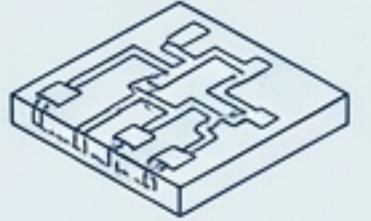


Planar Transformation

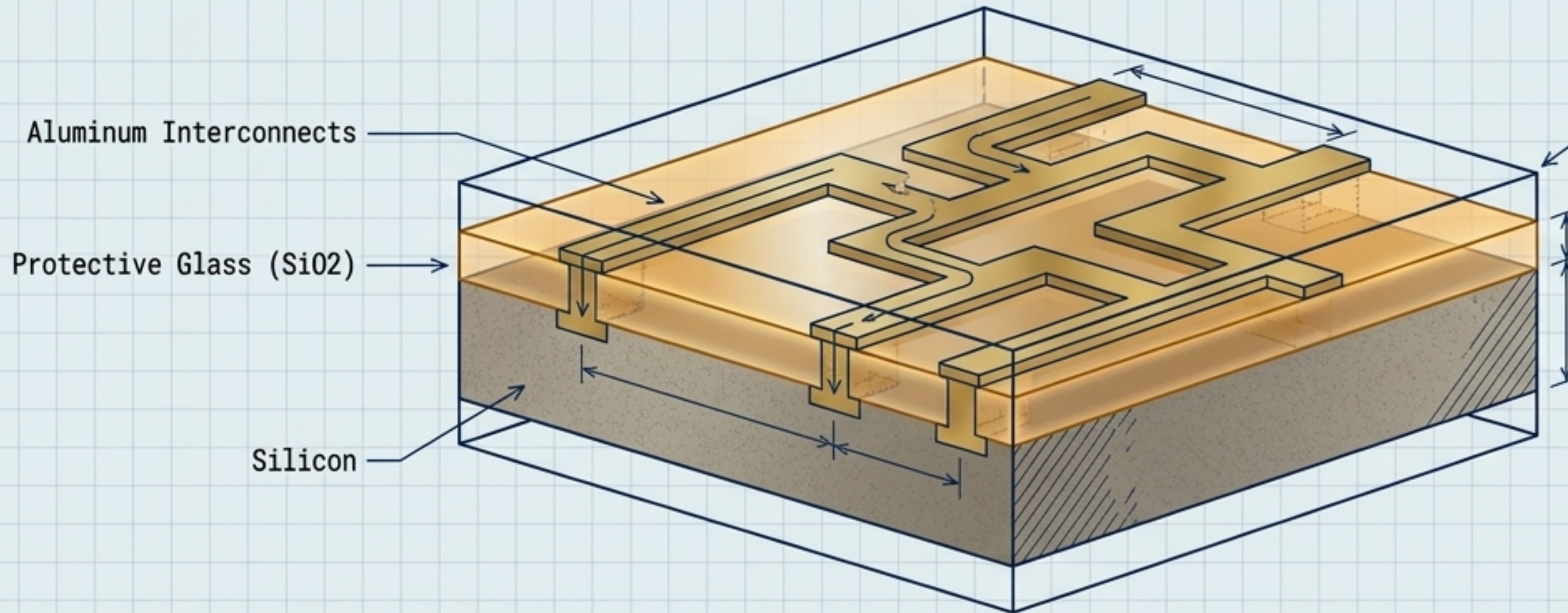
[Noyce's Integrated Circuit (1961)]

Depositing aluminum lines over Hoerni's glass to connect 4 transistors and 5 resistors on a single piece of silicon.

DATA: MOREN, Noyce, 1961

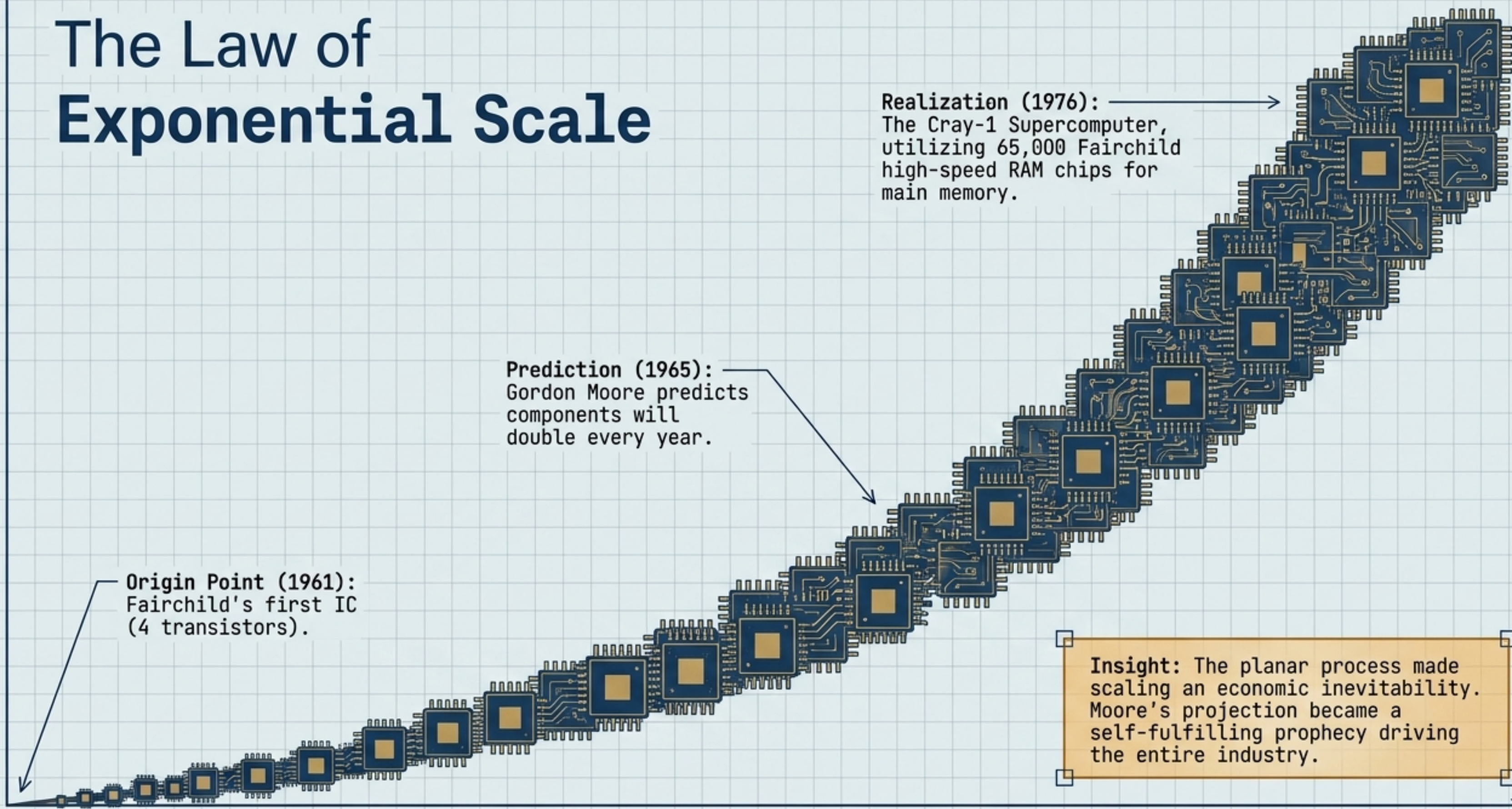


Monolithic Integration



SCHEMATIC VIEW: PLANAR PROCESS
LAYER STRUCTURE.
SCALE: NANOMETER TOLERANCE.
DATE: 1959-1961.
PATENT REF: HOERNI, NOYCE.

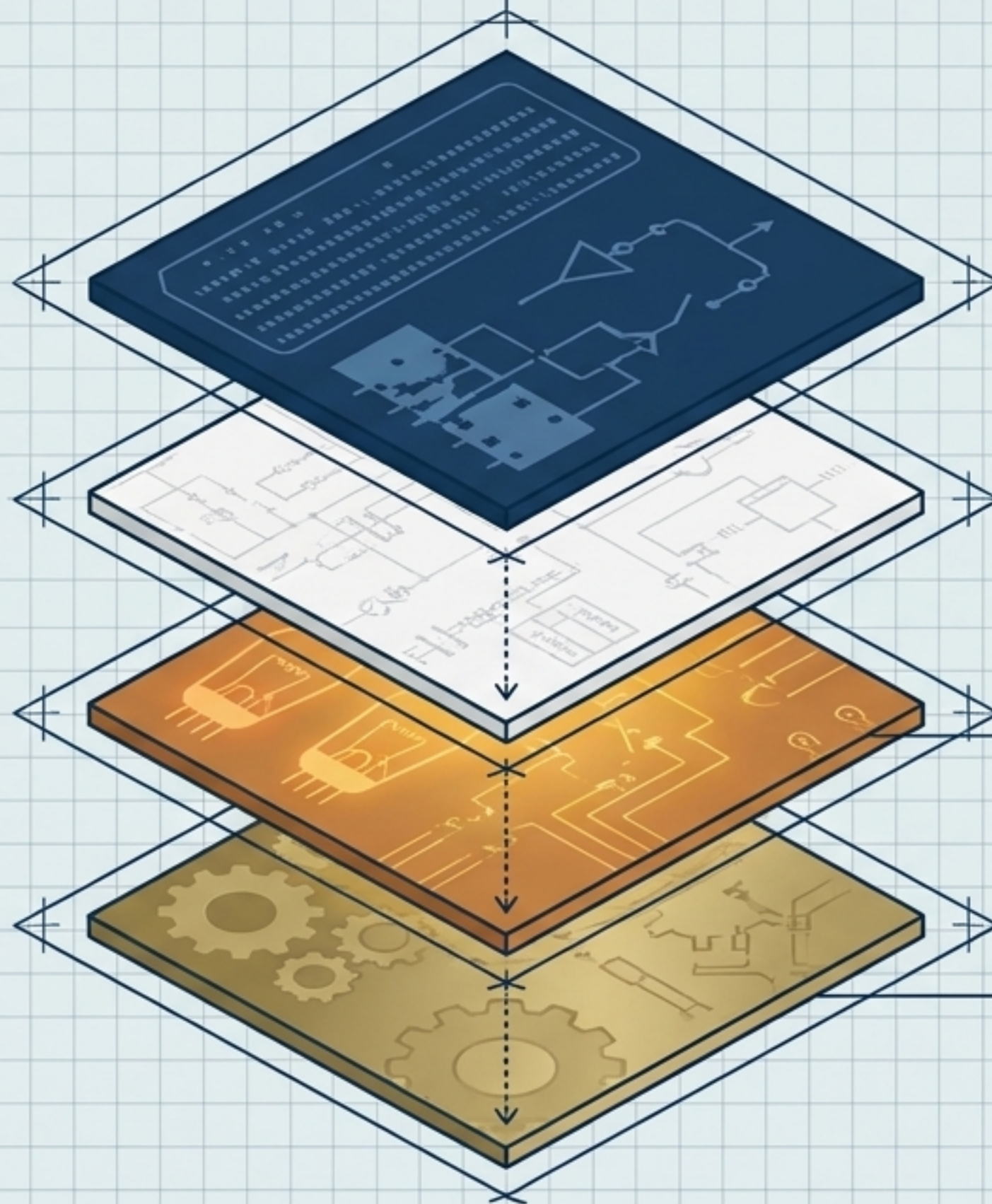
The Law of Exponential Scale



	Analytical Engine (1840s)	ENIAC (1946)	Transistor (1947)	Integrated Circuit (1961)
Base Component	Brass Gears	Vacuum Tubes	Germanium	Silicon & SiO ₂
Computing Paradigm	Mechanical	Electronic	Solid-State	Integrated
Programming Interface	Punch Cards	Rewiring / Switches	Assembly	Logic Simulators / EDA
Societal Perception	Theoretical	Awesome Super-Brain	Industrial Magic	Ubiquitous Law

The Stack Assembled

Lovelace's abstract logic is executed by ENIAC's electronic architecture, powered by Bell's solid-state switch, and infinitely scaled by Fairchild's printed silicon.



The Logic — Programs as Data / Conditional looping (Lovelace/Babbage).

The Machine — Electronic operational speed (ENIAC).

The Switch — Solid-state transistors (Bell Labs).

The Material — Silicon & planar manufacturing (Fairchild).

1843 (The Theory).

The Analytical Engine has no pretensions whatever to originate anything.

1946 (The Hype).

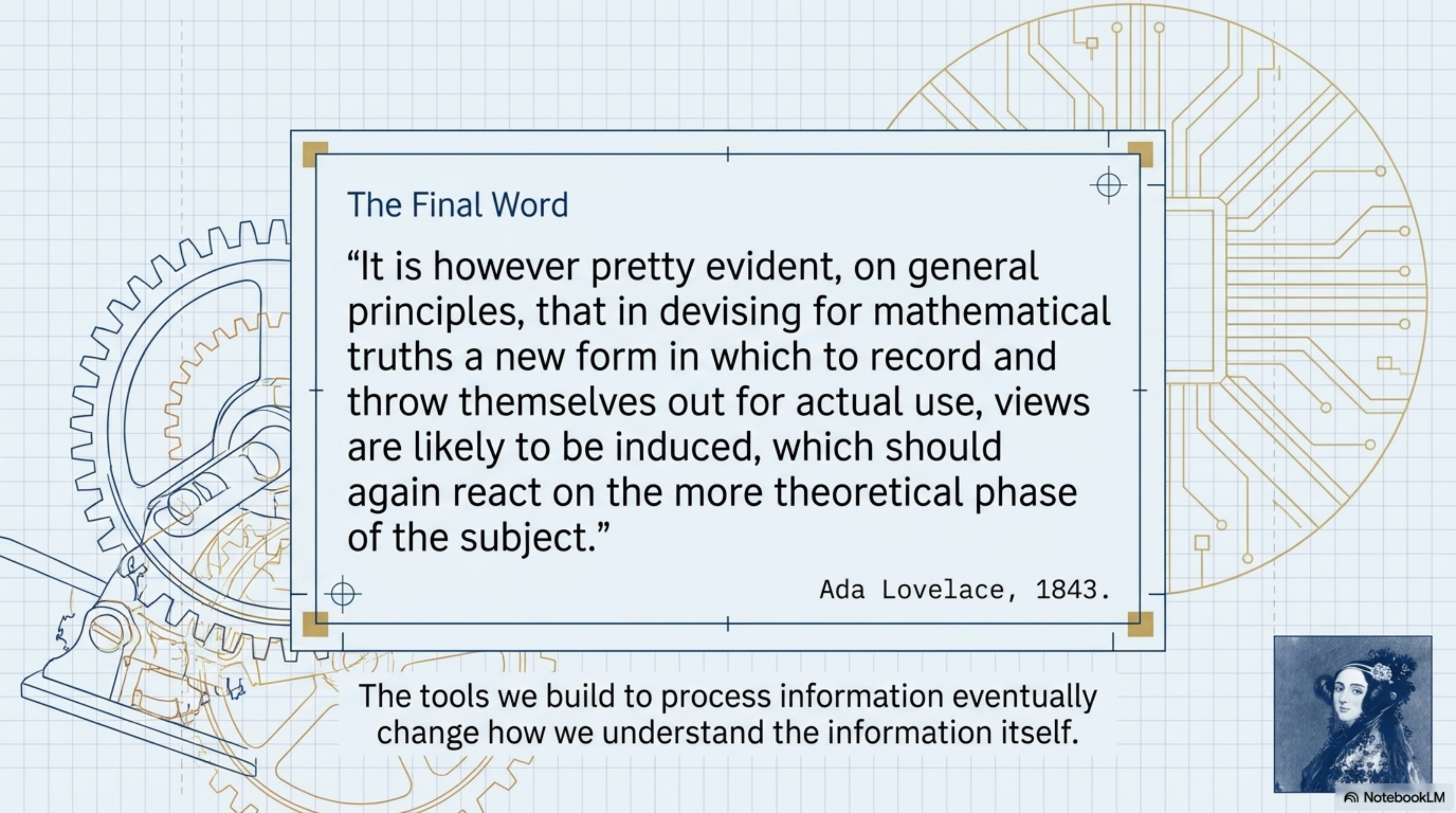
Media Reality: Super-Brain, Frankenstein, Mechanical Einstein.

Turing's Warning: Deprecates the notion that a machine is a complete substitute for the human brain.

1970s (The Disillusionment).

New York Times (1971): The computer has made a significant impact... though it has not been as useful as some of its supporters supposed.

The Pattern: Exaggerated metaphors consistently breed premature enthusiasm, followed by an inevitable perception crash.



The Final Word

“It is however pretty evident, on general principles, that in devising for mathematical truths a new form in which to record and throw themselves out for actual use, views are likely to be induced, which should again react on the more theoretical phase of the subject.”

Ada Lovelace, 1843.

The tools we build to process information eventually change how we understand the information itself.

